

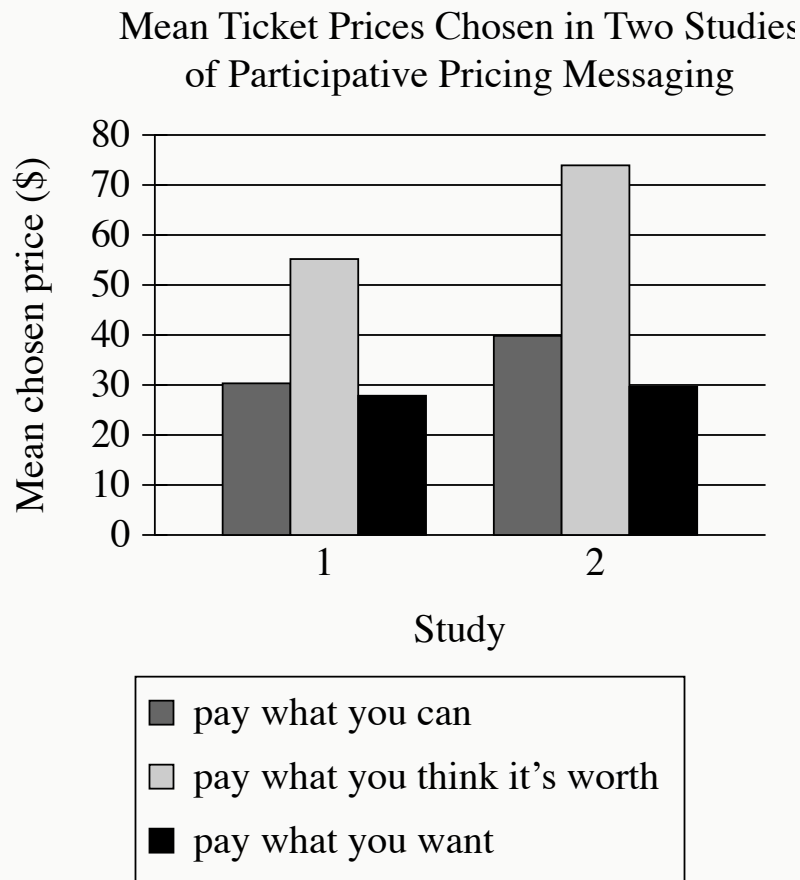
● HARD

● INFORMATION AND IDEAS

Mixed

10 questions · ~15 min

04

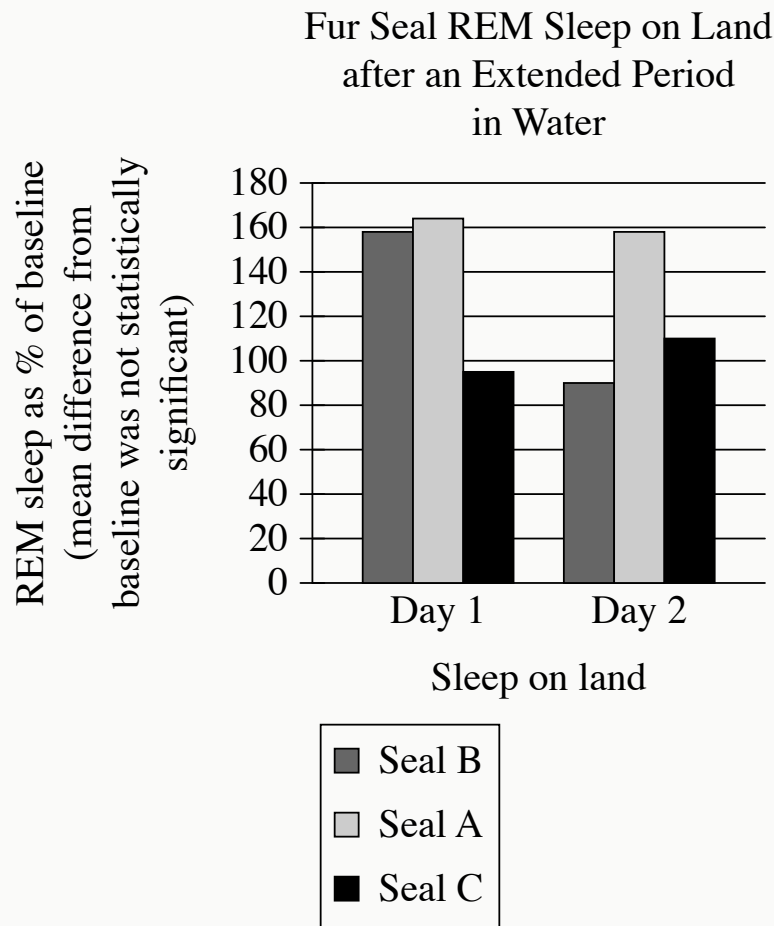


- For each data category, the following bars are shown:
 - pay what you can
 - pay what you think it's worth
 - pay what you want
- The Mean chosen price data for the 2 categories are as follows:
 - Study 1:
 - pay what you can: \$30.33
 - pay what you think it's worth: \$55.19
 - pay what you want: \$27.85
 - Study 2:
 - pay what you can: \$39.83
 - pay what you think it's worth: \$73.93
 - pay what you want: \$29.64

Participative pricing, in which purchasers choose the prices they pay for products, can enable sellers to capitalize on the heterogeneous values consumers assign to the same goods and services, but doing so requires careful messaging. Annie Peng Cui and Jennifer Wiggins recruited 171 participants (ages 18–60) online for an initial study and 83 students (ages 18–31) at a state university for a second study to test the effect of three different messages—“pay what you can,” “pay what you think it’s worth,” and “pay what you want”—on how much participants would pay for concert tickets. Their results illustrate both the heterogeneity of consumer valuations and how sellers can benefit by prompting consumers to consider their own valuations: _____blank

Which choice most effectively uses data from the graph to complete the text?

- A. the students tended to value the concert tickets more highly than did the more age-diverse group recruited online, but when considering what they could afford to pay, the students tended to choose a lower price than did the other group.
- B. in all three messaging conditions, the group of participants recruited online tended to choose lower prices than did the students, but both groups tended to choose prices closest to the actual cost of the tickets when prompted to consider the tickets’ value.
- C. the students tended to value the concert tickets more highly than did the more age-diverse group recruited online, but both groups tended to choose a higher price when considering the value of the tickets than when considering what they could afford or wanted to pay.
- D. within each group of participants, there was wide variation in the value that individuals assigned to the concert tickets, but the students tended to assign a higher value to the tickets than did the more age-diverse group recruited online.



Research suggests that REM sleep in animals is homeostatically regulated: animals compensate for periods of REM sleep deprivation by increasing subsequent REM sleep. When on land, fur seals get enough REM sleep, but during the weeks they're in the water, they get almost none. In a study of fur seals' sleep habits, researchers recorded the REM sleep (as a percentage of baseline) of fur seals once they had returned to land. They concluded that REM sleep may not be homeostatically regulated in fur seals, citing as evidence the fact that the seals in the study _____blank

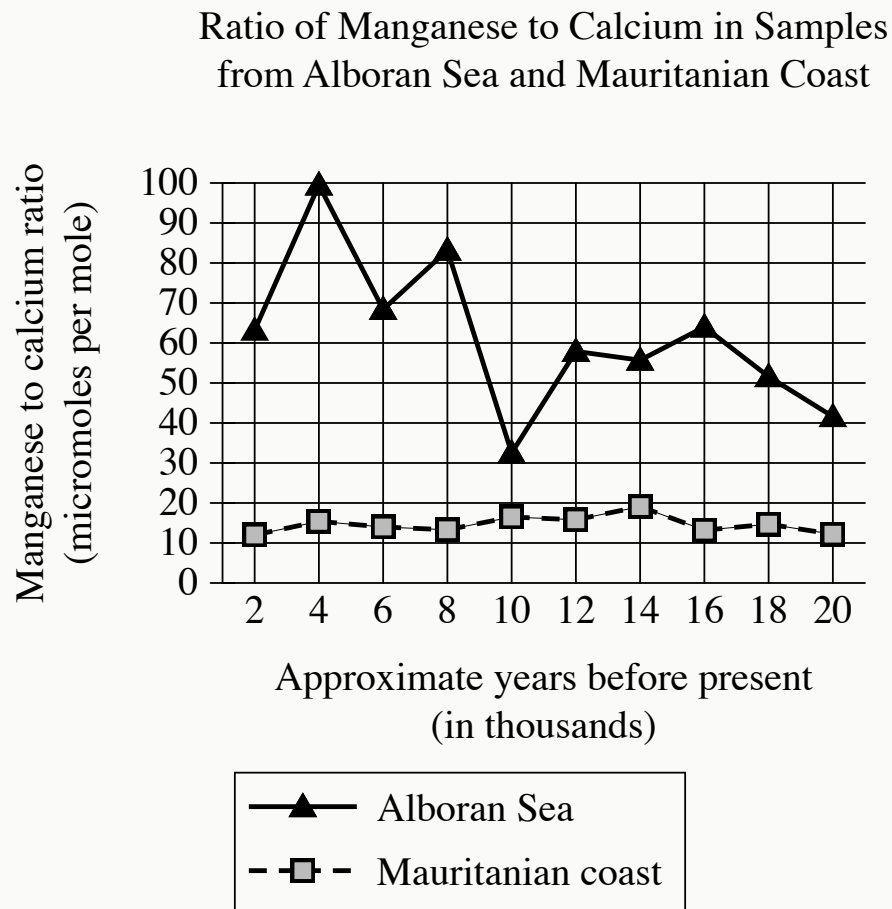
Which choice most effectively uses data from the graph to complete the text?

- A. didn't show significantly less REM sleep during the second day after returning to land than they did during the first day.
- B. showed no significant differences from one another in baseline levels of REM sleep.
- C. didn't consistently demonstrate a significant increase in REM sleep after their period of deprivation in the water.
- D. showed no significant difference between REM sleep after returning to land and REM sleep while in the water.

Studies of ocean wave breaking have predominantly focused on traveling waves (those propagating along the horizontal plane), so Mark McAllister et al. utilized a circular wave tank to produce and study spike waves, axisymmetric standing waves that can erupt vertically when traveling waves propagating in opposing directions intersect. Traveling waves break when wave steepness (height-to-length ratio) passes a critical threshold; breaking thus constrains wave height. McAllister et al. found that spike waves can exceed that constraint, as other factors than just steepness (e.g., jet stability and cavity shape) mediate spike-wave breaking.

Which choice best states the main idea of the text?

- A. Previous studies have suggested that steepness mediates breaking in traveling waves, but the study by McAllister et al. shows that jet stability and cavity shape may also influence breaking in such waves.
- B. The process of breaking limits the height of traveling waves, but the study by McAllister et al. suggests that spike waves can exceed those limits if their height-to-length ratio reaches a critical threshold.
- C. McAllister et al. suggest that spike waves can form when traveling waves propagating in opposing directions intersect and that spike waves tend to be higher than traveling waves.
- D. The study by McAllister et al. suggests that when traveling waves intersect in specific ways, the resulting wave may be higher than would be expected based on the properties of traveling waves.



- The following 2 lines are shown:
 - Alboran Sea
 - Mauritanian coast
- The Alboran Sea line:
 - Begins at 2 thousand, 63.2 micromoles per mole
 - Rises sharply to 4 thousand, 99.5 micromoles per mole
 - Falls sharply to 6 thousand, 68.4 micromoles per mole
 - Rises sharply to 8 thousand, 83.2 micromoles per mole
 - Falls sharply to 10 thousand, 32.5 micromoles per mole
 - Rises sharply to 12 thousand, 57.9 micromoles per mole
 - Falls gradually to 14 thousand, 55.7 micromoles per mole
 - Rises gradually to 16 thousand, 64 micromoles per mole

- Falls sharply to 18 thousand, 51.6 micromoles per mole
- Falls sharply to 20 thousand, 41.6 micromoles per mole
- The Mauritanian coast line:
 - Begins at 2 thousand, 12 micromoles per mole
 - Rises gradually to 4 thousand, 15.4 micromoles per mole
 - Falls gradually to 6 thousand, 14 micromoles per mole
 - Falls gradually to 8 thousand, 13.3 micromoles per mole
 - Rises gradually to 10 thousand, 16.5 micromoles per mole
 - Falls gradually to 12 thousand, 15.8 micromoles per mole
 - Rises gradually to 14 thousand, 19.1 micromoles per mole
 - Falls gradually to 16 thousand, 13.2 micromoles per mole
 - Rises gradually to 18 thousand, 14.7 micromoles per mole
 - Falls gradually to 20 thousand, 12.2 micromoles per mole

The population of the coral *Lophelia pertusa* declined significantly around 9,000 years ago in the Alboran Sea and around 11,000 years ago near the Mauritanian coast. Using the ratio of manganese to calcium, which inversely correlates with ocean oxygenation levels, marine scientist Rodrigo da Costa Portilho-Ramos and colleagues evaluated whether oxygenation played a role in the declines of *L. pertusa*. The researchers concluded that oxygenation may have been important in the Alboran Sea but not near the Mauritanian coast, since _____ blank

Which choice most effectively uses data from the graph to complete the statement?

- A. a substantial increase in oxygenation in the Alboran Sea corresponded with the local decline in *L. pertusa*, but the opposite relationship between oxygenation and *L. pertusa* was found near the Mauritanian coast.
- B. *L. pertusa* declined in the Alboran Sea during a period of substantial local decline in oxygenation, but *L. pertusa* declined near the Mauritanian coast during a period of little local change in oxygenation.
- C. oxygenation in the Alboran Sea was higher before the decline in *L. pertusa* than after the decline, whereas oxygenation near the Mauritanian coast was relatively low both before and after the decline in *L. pertusa*.

- D. oxygenation in the Alboran Sea tended to be substantially higher than oxygenation near the Mauritanian coast during the period studied.

While researching a topic, a student has taken the following notes:

- The Gullah are a group of African Americans who have lived in parts of the southeastern United States since the 18th century.
- Gullah culture is influenced by West African and Central African traditions.
- Louise Miller Cohen is a Gullah historian, storyteller, and preservationist.
- She founded the Gullah Museum of Hilton Head Island, South Carolina, in 2003.
- Vermelle Rodrigues is a Gullah historian, artist, and preservationist.
- She founded the Gullah Museum of Georgetown, South Carolina, in 2003.

The student wants to emphasize the duration and purpose of Cohen’s and Rodrigues’s work. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. At the Gullah Museums in Hilton Head Island and Georgetown, South Carolina, visitors can learn more about the Gullah people who have lived in the region for centuries.
- B. Louise Miller Cohen and Vermelle Rodrigues have worked to preserve the culture of the Gullah people, who have lived in the United States since the 18th century.
- C. Since 2003, Louise Miller Cohen and Vermelle Rodrigues have worked to preserve Gullah culture through their museums.
- D. Influenced by the traditions of West and Central Africa, Gullah culture developed in parts of the southeastern United States in the 18th century.

A turtle shell appears external to the animal, protecting its body like armor. _____ blank the shell is often incorrectly assumed to be an exoskeleton, a rigid outer casing like that of a crustacean or an insect, when in fact it is an endoskeleton, a part of the turtle's internal bone structure, more akin to a spine or a pair of ribs.

Which choice completes the text with the most logical transition?

- A. That being said,
- B. However,
- C. For instance,
- D. Hence,

In Marisol's 1968 sculpture *Mi Mama y Yo*, gone are the types of pop culture references that made the Parisian-born Venezuelan American artist a star at the height of the pop art movement. In _____ blank place is a far more personal subject: a sculptural depiction of the artist as a young girl with her mother.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. its
- B. they're
- C. their
- D. it's

During the decades-long movement to codify the rights of Latinos in the US, certain events were pivotal: the founding of social justice group the League of United Latin American Citizens in _____ blank *Katzenbach v. Morgan* court decision in 1966, which affirmed the rights of Latino voters, is another such event.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. 1929. For one, the
- B. 1929, for one, the
- C. 1929 for one, the
- D. 1929, for one. The

The following text is from John Dryden's 1697 translation of Virgil's poem the *Aeneid*, written in the first century BCE. Queen Dido has just heard the Trojan hero Aeneas recount a sad tale.

But anxious cares already seiz'd the queen:
She fed within her veins a flame unseen;
The hero's valor, acts, and birth inspire
Her soul with love, and fan the secret fire.
His words, his looks, imprinted in her heart,
Improve the passion, and increase the smart.

Which choice best states the main purpose of the text?

- A. To indicate that Dido is hiding her skepticism of Aeneas's account
- B. To emphasize that Dido has been deeply affected by Aeneas's story
- C. To imply that Dido engages both emotionally and intellectually with Aeneas
- D. To suggest that Dido is worried that Aeneas might discover a secret she is keeping from him

In Jane Austen’s novel *Mansfield Park*, an almost imperceptible smile from potential suitor Henry Crawford causes the protagonist Fanny Price to blush; her embarrassment grows when she suspects that he is aware of it. This moment—in which Fanny not only infers Henry’s mental state through his gestures, but also infers that *he* is drawing inferences about *her* mental state—illustrates what literary scholar George Butte calls “deep intersubjectivity,” a technique for representing interactions between consciousnesses through which Austen’s novels derive much of their social and psychological drama.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A. It states a claim about Austen’s skill at representing psychological complexity that is reinforced by an example presented in the following sentence.
- B. It advances an interpretation of an Austen protagonist who is contrasted with protagonists from other Austen novels cited in the following sentence.
- C. It describes a recurring theme in Austen’s novels that is the focus of a literary scholar’s analysis summarized in the following sentence.
- D. It provides a synopsis of an interaction in an Austen novel that illustrates a literary concept discussed in the following sentence.